

A NOTE ON NODALLY BOUND-PRESERVING FINITE ELEMENT METHODS FOR THE GINZBURG–LANDAU ENERGY

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ABSTRACT. We study conforming piecewise affine finite element approximations of minimizers of the reduced Ginzburg–Landau energy with a prescribed magnetic vector potential. Although continuous critical points satisfy the modulus bound $|u| \leq 1$, this property is not automatically guaranteed for their discrete counterparts. We enforce it nodally by minimizing over the constraint set $|v_h(x_i)| \leq 1 + \delta$, where $\delta \geq 0$. For quasi-isolated minimizers, we derive an error estimate in the κ -weighted H^1 -norm in terms of a constrained nonlinear Ritz projection. We also characterize this projection through an equivalent stabilized nonlinear variational problem. For the relaxed constraint $\delta > 0$ in two and three dimensions, explicit resolution conditions yield optimal-order L^2 approximation of the Ritz projection and an $\mathcal{O}(\kappa h)$ error bound for the discrete minimizer in the weighted energy norm. For the strict constraint $\delta = 0$ in one dimension, we adapt a weighted Aubin–Nitsche argument to the complex-valued modulus constraint and obtain an $\mathcal{O}(\kappa^2 h^{3/2-\varepsilon})$ L^2 -estimate. Under a corresponding resolution condition, the optimal $\mathcal{O}(\kappa h)$ energy-norm convergence rate is retained.

1. INTRODUCTION

In this work, we study discrete minimizers of the (reduced) Ginzburg–Landau (GL) energy

$$(1.1) \quad E(u) := \frac{1}{2} \int_{\Omega} \left| \frac{i}{\kappa} \nabla u + \mathbf{A}u \right|^2 + \frac{1}{2} (1 - |u|^2)^2 \, dx,$$

on some convex, bounded polyhedral domain $\Omega \subset \mathbb{R}^d$, $d = 1, 2, 3$, for a given vector potential. We use this as a model problem to provide a proof of concept for the full model

$$E(u, \mathbf{A}) := \frac{1}{2} \int_{\Omega} \left| \frac{i}{\kappa} \nabla u + \mathbf{A}u \right|^2 + \frac{1}{2} (1 - |u|^2)^2 + |\operatorname{curl} \mathbf{A} - \mathbf{h}_{\text{ext}}|^2 \, dx,$$

where $\mathbf{h}_{\text{ext}}$ instead of $\mathbf{A}$ is given, and the minimization has to be performed with respect to the pair $(u, \mathbf{A})$. This model serves as a macroscopic description of superconductivity first observed by Kamerlingh Onnes in [15], and in particular describes type-II superconductors and the formation of an Abrikosov vortex lattice [1]. In the usual GL interpretation, $|u|^2$ represents the density of so-called Cooper pairs with $|u| = 1$ being the superconducting state and $|u| = 0$ the normal state. In fact, the results in [10] and [18] show that any critical point of E satisfies $|u| \leq 1$.

However, to the best of our knowledge there is no result in the literature ensuring that this property is inherited in the discrete case. Our approach is to enforce a maximum bound directly in the minimization process while retaining quasi-optimality of the approximation. Our main tool is the nonlinear discretization proposed in [2, 4] invoking mass-lump-type stabilization terms for real-valued scalar reaction-(convection-)diffusion problems under pointwise constraints. The authors obtain optimal convergence in the associated energy norm. In our analysis, we additionally require improved convergence in L^2 which is not provided in those works. Recently, [3] recovered optimal-order L^2 convergence in one dimension by a modified Aubin–Nitsche trick. In this work, we adapt this to the nodal modulus bound constraint also in the (nonphysical) one-dimensional case. In higher dimension, we can still obtain optimal order of convergence under the relaxed constraint $|u_h| \leq 1 + \delta$ for some $\delta > 0$, but the resolution condition blows up for $\delta \rightarrow 0$. Extensions to higher dimensions for $\delta = 0$ are part of future investigations.

We now give a brief summary of the literature on the stationary Ginzburg–Landau problem. For the analysis of the problem, there are many important contributions by Sandier and Serfaty in a series of works [18–21]. The seminal works by Du, Gunzburger and Peterson [10, 11] were the

first to provide rigorous error bounds for the finite element discretization but with rates that are not explicit in the parameter κ . In [13] a spatial discretization based on a covolume method is suggested, and in [12] a finite volume discretization is used to solve the GL problem on the sphere. The first attempt for κ -explicit error estimates was made by one of the authors in [9], which showed, in particular, that the finite element error behaves asymptotically as well as the H^1 -best-approximation error. These findings were then generalized to discrete multiscale spaces and also the full model in [5] and [7]. More recently, in [6] the authors provided a fully explicit resolution condition also for higher-order polynomials on smooth domains, which was then extended in [14] to discrete multiscale spaces. For non-convex polyhedral domains, the results in [8] provide error bounds which are – at least in dimension two – explicit on the size of the angle of largest re-entrant corners. Even though there is an extensive literature on the time-dependent variant, we only mention [16] where the authors prove a discrete maximum-bound principle for the order parameter.

2. GINZBURG-LANDAU PROBLEM AND DISCRETE MINIMIZERS

2.1. Preliminaries and recap of results. For the mathematical framework, we require the following notation: For $2 \leq p \leq \infty$ we denote by $L^p = L^p(\Omega, \mathbb{C})$ the standard Lebesgue space of scalar complex-valued functions on Ω equipped with the usual norm $\|\cdot\|_{L^p}$. For $p = 2$ the Hilbert space L^2 is considered as a *real* Hilbert space with the inner product

$$(v, w) = \operatorname{Re} \int_{\Omega} v \bar{w} \, dx$$

where $\bar{w}$ denotes the complex conjugate of w . If no confusion can arise, we also write $\|v\| = \|v\|_{L^2}$. For any integer $k \geq 1$, we denote by $H^k(\Omega)$ the standard Sobolev space of complex-valued functions in $L^2(\Omega)$ whose weak derivatives up to order k also belong to $L^2(\Omega)$. In this work, we study the minimization problem: find $u \in H^1(\Omega)$ such that

$$(2.1) \quad E(u) = \min_{v \in H^1(\Omega)} E(v),$$

From now on, we assume that $\Omega \subset \mathbb{R}^d$, $d = 1, 2, 3$, is a convex polyhedral domain and that

$$\mathbf{A} \in L^\infty(\Omega) \quad \text{with} \quad \operatorname{div} \mathbf{A} = 0 \text{ in } \Omega \quad \text{and} \quad \mathbf{A} \cdot \nu|_{\partial\Omega} = 0.$$

Further, we introduce the scaled norms

$$\|v\|_{H_\kappa^1}^2 = \kappa^{-2} \|\nabla v\|^2 + \|v\|^2, \quad \|v\|_{H_\kappa^2}^2 = \kappa^{-4} \|D^2 v\|^2 + \kappa^{-2} \|\nabla v\|^2 + \|v\|^2,$$

where D^2 is the collection of all second derivatives. The following result is now well known in the literature; see, for example, [9].

Lemma 2.1. *There exists at least one minimizer $u \in H^2(\Omega)$ of (2.1) with $\nabla u \cdot \nu|_{\partial\Omega} = 0$. In addition, it satisfies*

$$|u| \leq 1 \text{ a.e. in } \Omega, \quad \|u\|_{H_\kappa^1} + \|u\|_{H_\kappa^2} + \kappa^{-1} \|\nabla u\|_{L^4} \lesssim 1, \quad \kappa^{-1} \|\nabla u\| \lesssim \|u\|_{L^2},$$

with constants independent of κ .

Remark 2.2. *The pointwise bound obtained in the above lemma is established for example in [18] and [10]. The approach outlined in [18] derives a semilinear elliptic PDE for the modulus $|u|^2$ and uses (real) maximum bound principles to obtain the result. However, this approach makes explicit use of the strong formulation and thus appears not feasible for the discrete analysis. On the other hand, the proof provided in [10] constructs special test functions in the weak formulation of the critical point equation. However, in the discrete case, the same choice is polluted by the interpolation.*

Next, we define the first Fréchet derivative of the energy functional $E'(u) = \partial_u E(u) : H^1 \rightarrow \mathbb{R}$, which is given by

$$\langle E'(u), v \rangle = \left(\frac{i}{\kappa} \nabla u + \mathbf{A}u, \frac{i}{\kappa} \nabla v + \mathbf{A}v \right) + (|u|^2 - 1)u, v, \quad v \in H^1(\Omega)$$

By the first-order optimality condition each minimizer u needs to be a critical point of E , i.e., $\langle E'(u), v \rangle = 0$ for all $v \in H^1$. This leads to the famous Ginzburg–Landau equation

$$\left(\frac{i}{\kappa} \nabla u + \mathbf{A}u, \frac{i}{\kappa} \nabla v + \mathbf{A}v \right) + (|u|^2 - 1)u, v = 0$$

for all $v \in H^1(\Omega)$. Further, we directly observe that the energy is invariant under the complex rotation $e^{i\varphi}$ for all $\varphi \in \mathbb{R}$, and thus we cannot assume uniqueness of the minimizer. To make this more precise, we introduce the second Fréchet derivative $E''(u) : H^1 \times H^1 \rightarrow \mathbb{R}$ defined by

$$\langle E''(u)w, v \rangle = \operatorname{Re} \int_{\Omega} \left(\frac{i}{\kappa} \nabla w + \mathbf{A}w \right) \left(\frac{i}{\kappa} \nabla v + \mathbf{A}v \right)^* + (|u|^2 - 1)wv^* + u^2 w^* v^* + |u|^2 wv^* \, dx.$$

An easy computation shows that iu is always in the kernel of $E''(u)$, and thus we retrieve uniqueness via the following definition, see [9, Def. 2.4].

Definition 2.3. *Let u be a minimizer of (2.1). We call u a quasi-isolated minimizer if*

$$\langle E'(u), v \rangle = 0 \text{ for all } v \in H^1(\Omega), \quad \text{and} \quad \langle E''(u)v, v \rangle > 0 \text{ for all } v \in H_{iu}^1 \setminus \{0\},$$

where $H_{iu}^1 = H^1(\Omega) \cap (iu)^\perp = \{v \in H^1(\Omega) \mid (v, iu) = 0\}$.

The key tool throughout the analysis is to work in the space H_{iu}^1 instead of all of $H^1(\Omega)$. We obtain the following result, see [5, Prop.2.2].

Lemma 2.4. *Let u be a quasi-isolated minimizer of (2.1) in the sense of Definition 2.3. Then, there exists $1 \lesssim \rho = \rho(\kappa)$ such that*

$$\langle E''(u)v, v \rangle \geq \rho(\kappa)^{-1} \|v\|_{H_{iu}^1}^2 \text{ for all } v \in H_{iu}^1,$$

and further

$$\langle E''(u)v, w \rangle \lesssim \|v\|_{H_{iu}^1} \|w\|_{H_{iu}^1} \text{ for all } v, w \in H_{iu}^1,$$

with a constant independent of κ .

We conclude the section by noting the identity

$$(2.2) \quad \langle E''(u)u, v \rangle = \langle E'(u), v \rangle + \operatorname{Re} \int_{\Omega} 2|u|^2 uv^* \, dx,$$

and introduce the inner product

$$(2.3) \quad a_{\kappa}(w, v) = \operatorname{Re} \int_{\Omega} \left(\frac{i}{\kappa} \nabla w + \mathbf{A}w \right) \left(\frac{i}{\kappa} \nabla v + \mathbf{A}v \right)^* \, dx + \beta \operatorname{Re} \int_{\Omega} wv^* \, dx$$

with $\beta = 1 + \|\mathbf{A}\|_{L^\infty}^2$ used throughout this work.

3. MINIMIZATION UNDER MBP CONSTRAINT

In order to have some more flexibility in the analysis, we do not set the constraint to 1, but also take some $\delta \geq 0$, and constrain to $1 + \delta$. We treat the $\delta = 0$ and $\delta > 0$ differently. Let $V_h \subset H^1(\Omega)$ be an N -dimensional finite element space with a shape-regular and quasi-uniform mesh $\mathcal{T}_h$ of mesh size h and nodal Lagrange basis functions φ_i , nodes $\mathbb{A} = \{x_i \mid i = 1, \dots, N\}$, and degrees of freedom $\xi_i \in \mathbb{C}$ for $i = 1, \dots, N$. For any $v_h = \sum_{i=1}^N \xi_i \varphi_i \in V_h$ let us define the corrected component v_h^+ and the error component v_h^- via

$$\xi_i^+ = \begin{cases} \xi_i & \text{if } |\xi_i| \leq 1 + \delta \\ (1 + \delta) \frac{\xi_i}{|\xi_i|} & \text{if } |\xi_i| > 1 + \delta, \end{cases} \quad v_h^+ = \sum_{i=1}^N \xi_i^+ \varphi_i, \quad v_h^- = \sum_{i=1}^N (\xi_i - \xi_i^+) \varphi_i.$$

Clearly, we have $v_h^+ = (v_h^+)^+$. We use the intersection of the discrete space with $(iu)^\perp$ to define $V_h^{iu} = V_h \cap (iu)^\perp$. For $g(x) = |x|^2 - (1 + \delta)^2$ we define the convex set of positive components,

$$(3.1) \quad \begin{aligned} V_h^+ &:= \{\varphi_h^+ \mid \varphi_h \in V_h\} = \{\varphi_h \in V_h \mid |\varphi_h| \leq 1 + \delta \text{ at all } x_i \in \mathbb{A}\} \\ &= \{\varphi_h \in V_h \mid g(\varphi_h) \leq 0 \text{ at all } x_i \in \mathbb{A}\} \end{aligned}$$

$$V_h^{+,iu} := V_h^+ \cap (iu)^\perp,$$

which is the space of admissible solutions if linear finite elements are considered.

We aim to find $E(v_h) \rightarrow \min$ in $V_h^{+,iu}$. The critical point criterion for (components of) minima in the interior of V_h is that the gradient $E'(v_h) = 0$ vanishes, while for critical points on the boundary, the gradients $E'(v_h)$ and $g'(v_h)$ need to be linearly dependent. That is, there is a $\lambda \in \mathbb{R}$, usually called Lagrange multiplier, such that $E'(v_h) = \lambda g'(v_h)$. If $\lambda > 0$, the critical point cannot only be a local minimum, while $\lambda < 0$ excludes a local maximum.

We now want to study the minimization problem: find $u_h \in V_h^+$

$$(3.2) \quad E(u_h) = \min_{v_h \in V_h^+} E(v_h),$$

Following [9], we may extract a sequence of minimizers as $h \rightarrow 0$ such that $u_h \in V_h^{+,iu}$ and $u_h \rightarrow u$ in $H^1(\Omega)$. For the rest of this work, we fix this sequence.

For the error analysis, a crucial tool is the constrained Ritz projection R_h for the inner product $a_\kappa(\cdot, \cdot)$ defined in (2.3) via

$$(3.3) \quad R_h u = \operatorname{argmin}_{v_h \in V_h^+ \cap (iu)^\perp} \frac{1}{2} \|u - v_h\|_{a_\kappa}^2.$$

We note that R_h is well-defined by the strict convexity of the functional and the feasible set. We note however that the operator is not linear. Nevertheless, we obtain the boundedness

$$\|R_h u\|_{a_\kappa} \leq \|u\|_{a_\kappa} + \|u - R_h u\|_{a_\kappa} \leq 2\|u\|_{a_\kappa},$$

where we used that $0 \in V_h^{+,iu}$. In addition, we immediately observe by the definition of R_h

$$\|u - R_h u\|_{a_\kappa} \leq \inf_{v_h \in V_h^+ \cap (iu)^\perp} \|u - v_h\|_{a_\kappa}.$$

With this, we can present our first main result for – at the moment – two independent minimizers.

Theorem 3.1. (a) Let $u \in H_\kappa^1 \setminus \{0\}$ be a quasi-isolated minimizer of (1.1) and $u_h \in V_h^{+,iu}$ a minimizer of (3.2). Then, we have the estimate

$$\|u - u_h\|_{H_\kappa^1} \leq \|u - R_h u\|_{H_\kappa^1} + \rho_\kappa(u)(2 + \beta)\|u - R_h u\|_{L^2} + \rho_\kappa(u)(3\|u - u_h\|_{L^4}^2 + \|u - u_h\|_{L^6}^3)$$

(b) In addition, let u_h be a sequence that converges to u in $H^1(\Omega)$. Then, for h sufficiently small it holds

$$\|u - u_h\|_{H_\kappa^1} \leq \|u - R_h u\|_{H_\kappa^1} + \rho_\kappa(u)(2 + \beta)\|u - R_h u\|_{L^2}$$

with constants independent of κ and h .

This result is proved in Section 4. The essential tool used in the proof is the variational inequality satisfied by the Ritz projection, which we establish below. However, no approximation properties are used so far, and the above statement holds independently of these results.

3.1. Main results on the Ritz projection. In order to conclude an error bound from Theorem 3.1, we now provide estimates on the Ritz projection. First, we consider the approximation in H_κ^1 , and obtain the following estimate. We note that at this stage this bound is only valid for the minimizer itself. The proof is presented in Section 5.

Theorem 3.2. Let $u \in H^2(\Omega)$ be a quasi-isolated minimizer of (1.1), and assume that $\kappa h \lesssim 1$ is sufficiently small, then

$$\|u - R_h u\|_{a_\kappa} \lesssim \kappa h$$

with a constant independent of κ and h .

However, in order to obtain convergence of possibly optimal order from Theorem 3.1, we need approximation of the Ritz projection also in L^2 . In this manuscript, we present two different types of results.

A. The relaxed case $\delta > 0$. The first result treats the relaxed case $\delta > 0$, but can deal with dimension $d = 2, 3$. Note that we could also state this result for the one-dimensional case, but as it is included in the second result, we refrain from doing so.

Theorem 3.3. *Let $d = 2$ or $d = 3$, $\delta > 0$, and h sufficiently small such that in particular for constants C_∞, C'_∞*

$$\begin{cases} C'_\infty \kappa^2 |\log h| h < \delta, & d = 2 \\ C'_\infty \kappa^2 h^{1/2} < \delta, & d = 3 \end{cases} \text{ are sufficiently small.}$$

(a) *Then, it holds*

$$\|u - R_h u\|_{L^2} \lesssim \kappa h \|u - R_h u\|_{a_\kappa} \lesssim (\kappa h)^2$$

with a constant independent of κ and h .

(b) *Further, if in addition $\rho(\kappa)\kappa h \lesssim 1$, we obtain the error bound*

$$\|u - u_h\|_{H^1_\kappa} \lesssim \kappa h$$

with a constant independent of κ and h .

The second part directly follows from Theorem 3.1, and the proof of the first part is presented in Section 6.

B. The strict case $\delta = 0$. The second result mimics the techniques of [3] for $\delta = 0$, but is only valid for $d = 1$. The proof is presented in Section 7.

Theorem 3.4. *Let $d = 1$ and $\delta = 0$. (a) Then, it holds*

$$\|u - R_h u\|_{L^2} \lesssim \kappa h^{1/2-\varepsilon} \|u - R_h u\|_a \lesssim \kappa^2 h^{3/2-\varepsilon}$$

with a constant independent of κ and h .

(b) *Further, if in addition h sufficiently small such that in particular $\rho(\kappa)\kappa h^{1/2-\varepsilon} \lesssim 1$, we obtain the error bound*

$$\|u - u_h\|_{H^1_\kappa} \lesssim \kappa h$$

with a constant independent of κ and h .

4. ERROR ANALYSIS

In this section, we follow the arguments from [9], but have to take care of the nonlinear Ritz projection. Therefore, we first derive the variational inequality for the Ritz projection, the critical point equation of the minimizers, and finally prove the first main result.

4.1. Variational inequality for the constrained Ritz projection. Recall the space $V_h^{+,iu}$ from (3.1) and we see that (3.3) clearly satisfies

$$R_h u = \operatorname{argmin}_{v_h \in V_h^+ \cap (iu)^\perp} J(v_h)$$

for $J(v_h) = \frac{1}{2} \|u - v_h\|_{a_\kappa}^2$. With this, we obtain the following well-known result, where we give the proof for convenience of the reader.

Lemma 4.1. *Let $R_h u$ be the unique minimizer of (3.3). Then, we have*

$$a_\kappa(u, v_h - R_h u) \leq a_\kappa(R_h u, v_h - R_h u)$$

for all $v_h \in V_h^{+,iu}$.

Below, we exploit this relation for $v_h = u_h$ in the form

$$(4.1) \quad a_\kappa(R_h u - u, R_h u - u_h) \leq 0$$

since $u_h \in V_h^{+,iu}$ is an admissible function by construction.

Proof of Lemma 4.1. We know that at the minimum it holds

$$\langle J'(R_h u)(v_h - R_h u) \rangle \geq 0$$

for all admissible $v_h \in V_h^{+,iu}$. We compute that

$$\begin{aligned} J(w_h + v_h) - J(w_h) &= \frac{1}{2} a_\kappa(u - w_h - v_h, u - w_h - v_h) - \frac{1}{2} a_\kappa(u - w_h, u - w_h) \\ &= \frac{1}{2} a_\kappa(v_h, v_h) - a_\kappa(u - w_h, v_h) \end{aligned}$$

such that

$$\langle J'(w_h)v_h \rangle = a_\kappa(w_h - u, v_h)$$

and hence, it holds

$$0 \leq \langle J'(R_h u)(v_h - R_h u) \rangle = a_\kappa(R_h u - u, v_h - R_h u)$$

which is equivalent to the assertion. $\square$

4.2. Derivation of the critical point equation. We express the minimization with the help of the functions

$$g_i(v_h) = \frac{1}{2} (|v_h(x_i)|^2 - (1 + \delta)^2), \quad i = 1, \dots, N,$$

equivalently as

$$\min_{v_h \in V_h} E(v_h) \quad \text{subject to} \quad g_i(v_h) \leq 0, \quad i = 1, \dots, N$$

Then, we obtain the Lagrangian function

$$F(v_h, \nu) = E(v_h) + \nu \cdot g$$

Now, let $u_h \in V_h^+$ be some minimizer of (3.2), then there exists some μ such that (u_h, μ) satisfy for all $\varphi_h \in V_h$

$$\begin{aligned} (4.2) \quad 0 &= E'(u_h)\varphi_h + \sum_{i=1}^N \mu_i \operatorname{Re}(u_h(x_i)\varphi_h(x_i)^*) \\ g(u_h) &\leq 0, \quad \mu \geq 0, \quad \mu \cdot g(u_h) = 0. \end{aligned}$$

4.3. Discrete error. We follow the strategy of [9, Section 5], and adapt the arguments to the constraint case. We decompose the error in its best-approximation part and the discrete part via

$$\|u - u_h\|_{H_\kappa^1} \leq \|u - R_h u\|_{H_\kappa^1} + \|R_h u - u_h\|_{H_\kappa^1}$$

The first part is part of the estimate in the theorem, and for the second part, we have by the coercivity of $E''(u)$ in Lemma 2.4

$$\begin{aligned} \|R_h u - u_h\|_{H_\kappa^1}^2 &\leq \rho_\kappa(u) \langle E''(u)(R_h u - u_h), (R_h u - u_h) \rangle \\ &= \rho_\kappa(u) \langle E''(u)(R_h u - u), (R_h u - u_h) \rangle + \rho_\kappa(u) \langle E''(u)(u - u_h), (R_h u - u_h) \rangle \\ &= \rho_\kappa(u) \varepsilon_h^{\text{lin}}(R_h u - u_h) + \rho_\kappa(u) \varepsilon_h^{\text{nonlin}}(R_h u - u_h) \end{aligned}$$

With this we can prove our first main result.

Proof of Theorem 3.1. We are only left to estimate the linear and nonlinear part. For the linear part, we compute

$$\begin{aligned} &\varepsilon_h^{\text{lin}}(R_h u - u_h) \\ &= a_\kappa(R_h u - u, R_h u - u_h) + \beta(R_h u - u, R_h u - u_h)_{L^2} \\ &\quad + \operatorname{Re} \int_\Omega (2|u|^2 - 1)(R_h u - u)(R_h u - u_h)^* + u^2(R_h u - u)^*(R_h u - u_h)^* \, dx \\ &\leq a_\kappa(R_h u - u, R_h u - u_h) + (2 + \beta) \|R_h u - u\|_{L^2} \|R_h u - u_h\|_{L^2}, \\ &\leq (2 + \beta) \|R_h u - u\|_{L^2} \|R_h u - u_h\|_{L^2}, \end{aligned}$$

where we used the property (4.1) for the last inequality. For the nonlinear part we employ the relation (2.2) to obtain

$$\begin{aligned}
& \varepsilon_h^{\text{nonlin}}(R_h u - u_h) \\
&= \langle E''(u)(u - u_h), (R_h u - u_h) \rangle \\
&= \langle E''(u)u, (R_h u - u_h) \rangle - \langle E''(u_h)u_h, (R_h u - u_h) \rangle + \langle E''(u_h)u_h, (R_h u - u_h) \rangle - \langle E''(u)u_h, (R_h u - u_h) \rangle \\
&= \text{Re} \int_{\Omega} 2|u|^2 u (R_h u - u_h)^* dx - \langle E'(u_h), (R_h u - u_h) \rangle - \text{Re} \int_{\Omega} 2|u_h|^2 u_h (R_h u - u_h)^* dx \\
&\quad + \text{Re} \int_{\Omega} (2|u_h|^2 - 1)u_h (R_h u - u_h)^* + u_h^2 u_h^* (R_h u - u_h)^* dx \\
&\quad - \text{Re} \int_{\Omega} (2|u|^2 - 1)u_h (R_h u - u_h)^* + u^2 u_h^* (R_h u - u_h)^* dx \\
&= \text{Re} \int_{\Omega} (2|u|^2 u - 2|u_h|^2 u_h + (2|u_h|^2 - 1)u_h + u_h^2 u_h^* - (2|u|^2 - 1)u_h + u^2 u_h^*) (R_h u - u_h)^* dx \\
&\quad - \langle E'(u_h), (R_h u - u_h) \rangle \\
&= \text{Re} \int_{\Omega} (2u|u - u_h|^2 + (u - u_h)^2 u^* - |u - u_h|^2 (u - u_h)) (R_h u - u_h)^* dx - \langle E'(u_h), (R_h u - u_h) \rangle.
\end{aligned}$$

Note that in the unconstrained case the last term would vanish. Here, we obtain from the critical point equation (4.2) that

$$\begin{aligned}
-\langle E'(u_h), (R_h u - u_h) \rangle &= \sum_{i=1}^M \mu_i \text{Re}(u_h(x_i)(R_h u - u_h)(x_i)^*) \\
&= \sum_{i=1}^M \mu_i (\text{Re}(u_h(x_i)R_h u(x_i)^*) - |u_h(x_i)|^2) \\
&\leq \sum_{i=1}^M \mu_i (|u_h(x_i)||R_h u(x_i)| - |u_h(x_i)|^2) = (1 + \delta) \sum_{i \in A} \mu_i (|R_h u(x_i)| - (1 + \delta)) \leq 0,
\end{aligned}$$

since by construction of the Ritz projection $|R_h u(x_i)| \leq 1 + \delta$ at all nodes, and where A contains the active set $\mu_i > 0$ where $|u_h(x_i)| = 1 + \delta$. Thus, we can conclude

$$\begin{aligned}
\varepsilon_h^{\text{nonlin}}(R_h u - u_h) &\leq \text{Re} \int_{\Omega} (2u|u - u_h|^2 + (u - u_h)^2 u^* - |u - u_h|^2 (u - u_h)) (R_h u - u_h)^* dx \\
&\leq (3\|u - u_h\|_{L^4}^2 + \|u - u_h\|_{L^6}^3) \|R_h u - u_h\|_{L^2}.
\end{aligned}$$

Adding both estimates gives the desired estimate for part (a). Absorption of the higher order terms gives part (b) for h sufficiently small. $\square$

5. APPROXIMATION OF THE CONSTRAINED RITZ PROJECTION IN H_{κ}^1 AND EQUIVALENT FORMULATION

In this section, we present the approximation property in the natural norm H_{κ}^1 by constructing a suitable upper bound, and second provide the framework for the L^2 estimate.

5.1. Approximation in H_{κ}^1 . Here, we give the proof of the H_{κ}^1 bound for R_h . We use a suitable projection combined with a phase shift.

Proof of Theorem 3.2. Let N_j be the patch around the node x_j , and define operator

$$\tilde{\pi}: \{w \in L^{\infty}(\Omega) \mid |w| \leq 1 \text{ a.e.}\} \rightarrow V_h^+, \quad (\tilde{\pi}w)(x_j) = \frac{1}{|N_j|} \int_{N_j} w(x) dx,$$

which is H^1 -stable and has optimal approximation properties in L^p . We may assume $u \neq 0$ without loss of generality.

(1) Next, we define the angle ψ_h such that $e^{i\psi_h}\tilde{\pi}u \perp iu$ and obtain using the bound on the tangent of ψ_h

$$|\psi_h| \lesssim \frac{|(\tilde{\pi}u, iu)|}{|(\tilde{\pi}u, u)|}$$

and thus

$$\begin{aligned} \|u - e^{i\psi_h}\tilde{\pi}u\|_{H_\kappa^1} &\leq \|u - \tilde{\pi}u\|_{H_\kappa^1} + \|(1 - e^{i\psi_h})\tilde{\pi}u\|_{H_\kappa^1} \\ &\lesssim \|u - \tilde{\pi}u\|_{H_\kappa^1} + \frac{|(\tilde{\pi}u, iu)|}{|(\tilde{\pi}u, u)|} \|\tilde{\pi}u\|_{H_\kappa^1} \end{aligned}$$

(2) We are left to estimate the fraction. We compute

$$\frac{|(\tilde{\pi}u, iu)|}{|(\tilde{\pi}u, u)|} \|\tilde{\pi}u\|_{H_\kappa^1} \lesssim \frac{\|\tilde{\pi}u - u\|_{L^2} \|u\|_{L^2}}{|(\tilde{\pi}u - u, u) + \|u\|_{L^2}^2|} \|\tilde{\pi}u\|_{H_\kappa^1}$$

and with $\|u\|_{H_\kappa^1} \lesssim \|u\|_{L^2}$ from Lemma 2.1 we have

$$\|\tilde{\pi}u - u\|_{L^2} \lesssim h\kappa \|u\|_{H_\kappa^1} \lesssim h\kappa \|u\|_{L^2},$$

such that we obtain for $\kappa h \lesssim 1$ and $\|\tilde{\pi}u\|_{H_\kappa^1} \lesssim \|u\|_{H_\kappa^1} \lesssim \|u\|_{L^2}$

$$\frac{\|\tilde{\pi}u - u\|_{L^2} \|u\|_{L^2}}{|(\tilde{\pi}u - u, u) + \|u\|_{L^2}^2|} \|\tilde{\pi}u\|_{H_\kappa^1} \lesssim \frac{\|\tilde{\pi}u - u\|_{L^2} \|u\|_{L^2}^2}{\|u\|_{L^2}^2 - h\kappa \|u\|_{L^2}^2} \lesssim \|\tilde{\pi}u - u\|_{L^2},$$

which gives the quasi-best-approximation $\|u - e^{i\psi_h}\tilde{\pi}u\|_{H_\kappa^1} \lesssim \|u - \tilde{\pi}u\|$. The regularity in Lemma 2.1 then gives the desired bound. $\square$

5.2. Check equivalence with constrained bilinear forms. We now aim at an equivalent formulation of the Ritz projection via a stabilized nonlinear variational formulation introduced in [4]. As this was done for a real-valued problem, we investigate the necessary modifications. We define the bilinear form and norm

$$s(v_h, w_h) = C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re} v_h(x_i) w_h(x_i)^*, \quad \|v_h\|_s^2 = s(v_h, v_h),$$

with a positive function α_h to be chosen below. With the definition of V_h^+ and $V_h^{+,iu}$ in (3.1), we can check the following properties, cf. [4, Lemma 3.1]

Lemma 5.1. *The bilinear form s satisfies*

$$\begin{aligned} s(v_h^- - w_h^-, v_h^+ - w_h^+) &\geq 0, \quad \text{for all } v_h, w_h \in V_h, \\ s(v_h^-, w_h - v_h^+) &\leq 0, \quad \text{for all } v_h \in V_h, w_h \in V_h^+. \end{aligned}$$

Proof. We only have to prove the first inequality, as it implies the second. Note that it is sufficient to check this for every nodal point x_i , since the weight α_h is positive, and we thus fix some x_i . We have to distinguish three cases.

(1) If $|v_h(x_i)|, |w_h(x_i)| \leq 1 + \delta$ then the negative parts both vanish, and the estimate holds.

(2) If $|v_h(x_i)|, |w_h(x_i)| > 1 + \delta$, we have

$$\begin{aligned} &= \operatorname{Re} \left(v_h(x_i) - v_h(x_i) \frac{1 + \delta}{|v_h(x_i)|} - w_h(x_i) + w_h(x_i) \frac{1 + \delta}{|w_h(x_i)|} \right) (1 + \delta) \left(\frac{v_h(x_i)}{|v_h(x_i)|} - \frac{w_h(x_i)}{|w_h(x_i)|} \right)^* \\ &= -(1 + \delta)^2 \left| \frac{v_h(x_i)}{|v_h(x_i)|} - \frac{w_h(x_i)}{|w_h(x_i)|} \right|^2 + (1 + \delta) \operatorname{Re} \left(v_h(x_i) - w_h(x_i) \right) \left(\frac{v_h(x_i)}{|v_h(x_i)|} - \frac{w_h(x_i)}{|w_h(x_i)|} \right)^* \geq 0 \end{aligned}$$

using the first part of Lemma A.1.

(3) If $|v_h(x_i)| > 1 + \delta$ and $|w_h(x_i)| \leq 1 + \delta$ (which is enough by symmetry), we have $w_h^-(x_i) = 0$ and hence

$$\begin{aligned} & \operatorname{Re} \left(v_h(x_i) - (1 + \delta) \frac{v_h(x_i)}{|v_h(x_i)|} \right) \left((1 + \delta) \frac{v_h(x_i)}{|v_h(x_i)|} - w_h(x_i) \right)^* \\ &= - \left| (1 + \delta) \frac{v_h(x_i)}{|v_h(x_i)|} - w_h(x_i) \right|^2 + \operatorname{Re} \left(v_h(x_i) - w_h(x_i) \right) \left((1 + \delta) \frac{v_h(x_i)}{|v_h(x_i)|} - w_h(x_i) \right)^* \geq 0 \end{aligned}$$

using the second part of Lemma A.1. $\square$

Further, we have to show that the a_κ -norm can be controlled with the s -bilinear form on the space V_h . This is given by the following result, cf eq. (3.8) in [4]. We denote by $h(x_i)$ the average mesh size around the point x_i .

Lemma 5.2. *Define the nodal weight function*

$$\alpha_h(x_i) = h(x_i)^d (\kappa^{-2} h(x_i)^{-2} + \|\mathbf{A}\|_{L^\infty} \kappa^{-1} h(x_i)^{-1} + \|\mathbf{A}\|_{L^\infty}^2 + \beta).$$

Then, there is a constant C such that

$$\|v_h\|_{a_\kappa}^2 \leq \frac{C}{C_{stab}} \|v_h\|_s^2$$

where C only depends on the shape-regularity of the mesh.

Proof. We denote by Ω_i the union of all elements touching x_i . Then

$$\begin{aligned} & \operatorname{Re} \int_{\Omega_i} \left| \frac{i}{\kappa} \nabla v_h + \mathbf{A} v_h \right|^2 + \beta |v_h|^2 dx \\ &= \operatorname{Re} \int_{\Omega_i} \left| \frac{1}{\kappa} \nabla v_h \right|^2 + (|\mathbf{A}|^2 + \beta) |v_h|^2 dx + 2 \operatorname{Re} \int_{\Omega_i} \left(\frac{i}{\kappa} \nabla v_h \right) \cdot (\mathbf{A} v_h)^* \\ &\lesssim (\kappa^{-2} h(x_i)^{-2} + (\|\mathbf{A}\|_{L^\infty}^2 + \beta) + \|\mathbf{A}\|_{L^\infty} \kappa^{-1} h(x_i)^{-1}) \|v_h\|_{L^2(\Omega_i)}^2 \end{aligned}$$

with a constant C that only depends on the shape regularity. Using the equivalence to the mass lumped inner product and then summing up over all Ω_i gives the claim. $\square$

This allows us to define the following (nonlinear) form

$$(5.1) \quad a_h(u_h; v_h) = a_\kappa(u_h^+, v_h) + s(u_h^-, v_h)$$

and with this the (nonlinear) problem: Find $u_h \in V_h^{iu}$ such that

$$(5.2) \quad a_h(u_h; v_h) = a_\kappa(u, v_h), \quad \text{for all } v_h \in V_h^{iu}.$$

Theorem 5.3. *The problem (5.2) has a unique solution $u_h \in V_h^{iu}$, and u_h^+ coincides with the minimizer of (3.3), i.e., $u_h^+ = R_h u$.*

Proof. We may proceed along the lines of [4, Theorem 3.2] to obtain a unique solution of (5.2). Note that all steps remain valid due to our choice of taking real inner product of the complex Hilbert space. By [4, Theorem 3.5], we can derive that u_h^+ satisfies the variational inequality of Lemma 4.1, and by uniqueness we obtain the equivalence of the solutions. $\square$

Let us emphasize that at no point, we require the computation of R_h neither by minimization nor by the variational inequality nor the nonlinear equation. It serves solely as a theoretical tool in the error analysis.

6. APPROXIMATION IN L^2 , THE CASE $\delta > 0$

In this section, we provide the proof of Theorem 3.3. The main idea is to use the relaxation by $\delta > 0$ to set up a bootstrapping argument. We first show that the Ritz projection is bounded in L^∞ by something strictly smaller than $1 + \delta$, which means that there is no active node for the constraint.

Lemma 6.1. *Let $u \in H^2(\Omega)$ be a minimizer of (1.1).*

(a) *If $d = 2$, then for $C_\infty \kappa^2 |\log h| h < \delta$, it holds*

$$\|R_h u\|_{L^\infty} < 1 + \delta.$$

(b) *If $d = 3$, then for $C'_\infty \kappa^2 h^{1/2} < \delta$, it holds*

$$\|R_h u\|_{L^\infty} < 1 + \delta.$$

Note that this constraint is in two dimensions comparable to the resolution condition in [6, Theorem 3.2], whereas there even appears the $\rho_\kappa(u)$ as a factor. If $\rho_\kappa(u)$ would grow at least like $\kappa^{3/2}$, then the resolution condition in the three dimensional case implies our condition.

Proof of Lemma 6.1. Let for the moment $\hat{u}_h \in V_h$ be arbitrary. Then, we estimate

$$\begin{aligned} \|u - R_h u\|_{L^\infty} &\leq \|u - \hat{u}_h\|_{L^\infty} + \|\hat{u}_h - R_h u\|_{L^\infty} \\ &\lesssim \|u - \hat{u}_h\|_{L^\infty} + h^{-d/p} \|\hat{u}_h - R_h u\|_{L^p}, \end{aligned}$$

and we will employ the different Sobolev embedding for the different spatial dimensions.

(a) For $d = 2$, we have by Lemma B.1

$$\begin{aligned} h^{-d/p} \|\hat{u}_h - R_h u\|_{L^p} &\leq h^{-2/p} \sqrt{p} \|\hat{u}_h - R_h u\|_{H^1} \\ &\lesssim h^{-2/p} \sqrt{p} \|\hat{u}_h - u\|_{H^1} + h^{-2/p} \sqrt{p} \kappa \|u - R_h u\|_{H_\kappa^1} \end{aligned}$$

Choosing $p \sim |\log h|$ yields that $h^{-2/p} \sqrt{p} \sim |\log h|^{1/2}$, and $\hat{u}_h = I_h u$, the estimate from Theorem 3.2, and the fact that $\kappa^{-2} \|u\|_{H^2} \lesssim 1$, we obtain

$$\|u - R_h u\|_{L^\infty} \lesssim h \|u\|_{H^2} + |\log h| (h \|u\|_{H^2} + \kappa h) \leq C_\infty \kappa^2 |\log h| h$$

and in particular for $C_\infty \kappa^2 |\log h| h^s < \delta$, it holds

$$\|R_h u\|_{L^\infty} \leq \|u\|_{L^\infty} + \|u - R_h u\|_{L^\infty} < 1 + \delta,$$

which gives the claim. $\square$

(b) For $d = 3$, we choose $p = 6$ to obtain

$$h^{-3/6} \|\hat{u}_h - R_h u\|_{L^6} \lesssim h^{-1/2} \|\hat{u}_h - u\|_{H^1} + h^{-1/2} \kappa \|u - R_h u\|_{H_\kappa^1} \lesssim h^{1/2} \kappa^2$$

and thus with some $C'_\infty > 0$, we obtain for $C'_\infty \kappa^2 h^{1/2} < \delta$ the bound

$$\|R_h u\|_{L^\infty} \leq \|u\|_{L^\infty} + \|u - R_h u\|_{L^\infty} < 1 + \delta,$$

which gives the second claim. $\square$

Now looking at the formulation (5.1), we see that the error component of $R_h u$ vanishes, and in fact satisfies instead of (5.2) the linear version

$$a_\kappa(R_h u, v_h) = a_\kappa(u, v_h), \quad \text{for all } v_h \in V_h^{iu}.$$

In particular, we recover the orthogonality $a_\kappa(R_h u - u, v_h) = 0$ for all $v_h \in V_h^{iu}$.

Proof of Theorem 3.3. We follow the standard Aubin-Nitsche trick. Let $w \in H_{iu}^1$ be the unique solution to

$$a_\kappa(w, \varphi) = (R_h u - u, \varphi)$$

for all $\varphi \in H_{iu}^1$. Hence, we obtain since $R_h u - u \in H_{iu}^1$ and by the orthogonality for all $w_h \in V_h^{iu}$

$$\|R_h u - u\|^2 = a_\kappa(w, R_h u - u) \leq \|w - w_h\|_{a_\kappa} \|R_h u - u\|_{a_\kappa}$$

the regularity results and best approximation in [9], then give the two claimed estimates. $\square$

7. APPROXIMATION IN L^2 , THE CASE $\delta = 0$

In this section, we provide the proof of Theorem 3.4. The main effort consists of translating the positive constraint into the modulus constraint. However, we have to make a modification in the orthogonality constraint, i.e. instead of looking for a discrete minimizer in a subset of $(iu)^\perp$, we use a subspace of $(I_h iu)^\perp$, assuming that $I_h u \neq 0$, and comment on the necessary changes in Section 7.1. We therefore study the problem as in [3] in the form

$$a_\kappa(\phi, v) = (\mu\phi, v) + (\eta\phi', v) + \gamma(\phi', v'), \quad \text{with } (\phi, v) = \operatorname{Re} \int_\Omega \phi v^* dx$$

with possibly space dependent μ, η, γ . We later have the scaling

$$\mu \sim 1, \quad \eta \sim \kappa^{-1}, \quad \gamma \sim \kappa^{-2}$$

We also assume that there is a constant such that

$$\|\phi\|_\mu \leq C \|\phi\|_{a_\kappa}$$

We study

$$a_\kappa(u, v) = f(v), \quad \text{for all } v \in V_0 = \{v \in H^1(\Omega) \mid (I_h v, I_h iu) = 0\},$$

and we assume for the exact solution

$$|u(x)| \leq 1 \text{ almost everywhere in } \Omega.$$

The bilinear form s is an adaptation of the above construction with $\kappa = 1$ and $\mathbf{A} = \text{const.}$

Choice of the weights: We define the complex weights

$$\omega_j = -\frac{u_h(x_j)^-}{u(x_j) - u_h(x_j)^+}$$

with the convention $0/0 = 0$, and observe the following.

Lemma 7.1. *Either, we have $u(x_j) = u_h(x_j)^+$ or the weights satisfy*

$$\operatorname{Re} \omega_j \geq 0$$

for all j .

Proof. If $|u_h(x_j)| \leq 1$ the weight is zero. Hence, we only have to consider $|u_h(x_j)| > 1$. Thus, by positivity of the absolute value and the minus sign in front of the weights, we obtain

$$\begin{aligned} 0 \leq \operatorname{Re} \omega_j &\iff 0 \geq \operatorname{Re}(u_h(x_j)^-(u(x_j) - u_h(x_j)^+)^*) \\ &= \operatorname{Re}\left((u_h(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|})(u(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|})^*\right) \\ &= -|u_h(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|}|^2 + \operatorname{Re}\left((u_h(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|})(u(x_j) - u_h(x_j))^*\right) \end{aligned}$$

and it is sufficient to show that the second part is non-positive. We denote by $c = \operatorname{Re} u_h(x_j)u(x_j)^*$ with $|c| \leq |u_h(x_j)|$, and neglect the point evaluation in the following:

$$\begin{aligned} \operatorname{Re}\left((u_h - \frac{u_h}{|u_h|})(u - u_h)^*\right) &= c - \frac{c}{|u_h|} - |u_h|^2 + |u_h| \\ &= \frac{c}{|u_h|^2}(|u_h|^2 - |u_h|) - |u_h|^2 + |u_h| \\ &= \left(\frac{c}{|u_h|^2} - 1\right)(|u_h|^2 - |u_h|) \leq 0 \end{aligned}$$

since

$$\frac{c}{|u_h|^2} - 1 \leq 0 \quad \text{and} \quad |u_h|^2 - |u_h| \geq 0,$$

which gives the claim. $\square$

We define

$$\begin{aligned} J_0 &= \{x_j \mid u_h^+(x_j) = u(x_j) \text{ and } u_h^-(x_j) \neq 0\} \\ \mathcal{H}_0 &= \{v \in V_0 \mid \operatorname{Re}(u_h(x_j)v(x_j)^*) = 0, \text{ for all } j \in J_0\} \end{aligned}$$

and note that $\omega_j = \infty$ if and only if $j \in J_0$, but in the following these weights will only be multiplied by zero. Further, we mention that for $j \in J_0$ the difference $u(x_j) - u_h^+(x_j) = 0$ and thus the error $u(x_j) - u_h^+(x_j) \in \mathcal{H}_0$ is a valid test function later on.

With the weights, we define for $\phi, v \in \mathcal{H}_0$ the stabilization

$$\tilde{s}(\phi, v) = C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re}(\omega_i \phi(x_i) v(x_i)^*),$$

which directly gives by Lemma 7.1

$$\tilde{s}(\phi, \phi) = C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re} \omega_i |\phi(x_i)|^2 \geq 0.$$

For the duality argument, we require the identity

Lemma 7.2. *It holds the equality*

$$\tilde{s}(u - u_h^+, v) = -s(u_h^-, v),$$

for all $v \in \mathcal{H}_0$.

Proof. We compute the sum and note that whenever $i \notin J_0$ it holds $\omega_i(u(x_i) - u_h^+(x_i)) = -u_h(x_i)^-$ and thus

$$\begin{aligned} \tilde{s}(u - u_h^+, v) + s(u_h^-, v) &= C_{\text{stab}} \sum_{i=1}^N \alpha_h(x_i) \operatorname{Re}(\omega_i(u(x_i) - u_h^+(x_i)) + u_h^-(x_i)) v(x_i)^* \\ &= C_{\text{stab}} \sum_{i \in J_0}^N \alpha_h(x_i) \operatorname{Re} u_h^-(x_i) v(x_i)^* \\ &= C_{\text{stab}} \sum_{i \in J_0}^N \alpha_h(x_i) \operatorname{Re}(u_h(x_i) v(x_i)^*) (1 - |u_h(x_i)|^{-1}) = 0 \end{aligned}$$

by the definition of $\mathcal{H}_0$. $\square$

Lemma 7.3. *Let $\vartheta \in \mathcal{H}_0$ and $\omega_j \in \mathbb{C}$ some weights. Then, there exists $\vartheta_h \in V_h \cap \mathcal{H}_0$ such that*

$$\gamma(w', \vartheta_h') = -C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re}(\omega_i w(x_i) \vartheta(x_i)^*), \text{ for all } w \in \mathcal{H}_0.$$

Proof. Since we have Neumann conditions, we eliminate the constants to obtain uniqueness. We again proceed as in [3, Lemma 3.3], and let ϑ_h be the solution, when we replace $w \in \mathcal{H}_0$ by $w \in \mathcal{H}_0 \cap V_h$. In particular, we may insert $I_h w$ for all $w \in \mathcal{H}_0$ and only have to show

$$\operatorname{Re}(w' - (I_h w)', \vartheta_h') = 0$$

however, on each sub interval using integration by parts, and the fact that ϑ_h' is constant on the sub interval

$$\operatorname{Re} \int_{x_i}^{x_{i+1}} (w' - (I_h w)') \vartheta_h' dx = [(w - I_h w) \vartheta_h']_{x_i}^{x_{i+1}} = 0$$

as the interpolation $w(x_j) - I_h w(x_j) = 0$. $\square$

We consider the adjoint problem for $\psi \in L^2$ find $\vartheta \in \mathcal{H}_0$ such that

$$\tilde{a}_\kappa(w, \vartheta) + \tilde{s}(w, \vartheta) = (\mu w, \psi)$$

for all $w \in \mathcal{H}_0$, where $\tilde{a}_\kappa$ is the same as a and the derivative of the mixed term is on the other side.

Lemma 7.4 (Lemma 3.2 in [3]). *There is a unique solution $\vartheta \in \mathcal{H}_0$ and satisfies*

$$\|\vartheta\|_\mu + \|\vartheta\|_{\tilde{a}_\kappa} \lesssim \|\psi\|_\mu$$

with constants independent of the coefficients.

Proof. Existence and uniqueness are clear, and we only insert

$$\|\vartheta\|_\mu \lesssim \tilde{a}_\kappa(\vartheta, \vartheta) \leq \tilde{a}_\kappa(\vartheta, \vartheta) + \tilde{s}(\vartheta, \vartheta) = (\mu \vartheta, \psi) \leq \|\vartheta\|_\mu \|\psi\|_\mu$$

and hence the claims. $\square$

Lemma 7.5 (Lemma 3.3 in [3]). *Consider $\psi \in L^2$ and let $\vartheta \in \mathcal{H}_0$ solve the adjoint problem. Then, $\vartheta_2 \in H^2(\mathcal{T}_h)$, and satisfies*

$$\gamma \|\vartheta_2\|_{H^{3/2-\varepsilon}(\mathcal{T}_h)} \lesssim (\mu^{1/2} + \eta) \|\psi\|_\mu$$

Proof. We define $\vartheta_2 = \vartheta - \vartheta_h$. Then, we observe that

$$\begin{aligned} \gamma(w', \vartheta'_2) &= -(\mu w, \vartheta) - (\eta w, \vartheta') - \tilde{s}(w, \vartheta) + (\mu w, \psi) + C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re}(\omega_i w(x_i) \vartheta(x_i)^*) \\ &= -(w, \mu \vartheta + \eta \vartheta' + \mu \psi) \end{aligned}$$

We denote by $f = \mu \vartheta + \eta \vartheta' + \mu \psi \in L^2$, and show regularity of ϑ_2 which satisfies

$$\gamma(w', \vartheta'_2) = (w, f) \text{ for all } w \in \mathcal{H}_0.$$

To obtain H^2 , we write on each subinterval I^ℓ : $w = \hat{w} + \alpha(I_h i u) \in H^1(I^\ell)$ with

$$\alpha = \frac{(I_h w, I_h i u)}{\|I_h u\|^2}, \quad |\alpha| \leq \frac{\|I_h w\|}{\|I_h u\|} \lesssim \frac{\|w\|_{H^{1/2+\varepsilon}}}{\|I_h u\|}$$

with $\hat{w} \in \mathcal{H}_0$. In addition, using the bounds on u from Lemma 2.1, we conclude further for $\kappa h \lesssim 1$

$$\|I_h u\| \geq (1 - c\kappa h) \|u\| \geq C(1 - c\kappa h) \|u\|_{H_\kappa^1}.$$

Then for all $w \in H^1(I^\ell)$

$$\gamma(w', \vartheta'_2) = \gamma(\hat{w}', \vartheta'_2) + \gamma\alpha(iu', \vartheta'_2) = (\hat{w}, f) + \gamma\alpha(iu', \vartheta'_2) = (w, f) - \alpha(iu, f) + \gamma\alpha(iu', \vartheta'_2)$$

where one can show as in Lemma 2.8 in [9] that $|\gamma\alpha(iu', \vartheta'_2)| \lesssim \|\alpha iu\|_{L^2} \|f\|_{L^2}$ and by orthogonality $\|\alpha iu\|_{L^2} \leq \|w\|_{L^2}$ such that

$$\gamma(w', \vartheta'_2) = (w, \tilde{f}) \text{ for all } w \in H^1(I_\ell)$$

with uniformly bounded $\tilde{f} \in H^{-1/2-\varepsilon}$, and the local $H^{3/2-\varepsilon}$ -regularity is restored. $\square$

We can now conclude along the lines of Theorem 3.4 in [3].

Theorem 7.6. *Under the assumptions of the preceding results, we obtain the bound*

$$\begin{aligned} \|u - R_h u\|_\mu &\lesssim \left(\mu^{1/2} \gamma^{-1} h^{3/2-\varepsilon} \|u - R_h u\|_\mu + \eta \|u' - (R_h u)'\| h^{1/2-\varepsilon} \right. \\ &\quad \left. + \gamma^{-1/2} h^{1/2-\varepsilon} \gamma^{1/2} \|u' - (R_h u)'\| \right) (\mu^{1/2} + \eta) \end{aligned}$$

with constants independent of the coefficients and h .

Inserting now the values of the coefficients, we arrive at

$$\|u - R_h u\| \lesssim \kappa^2 h^{3/2-\varepsilon} \|u - R_h u\| + \kappa h^{1/2-\varepsilon} \|u - R_h u\|_{a_\kappa}$$

and for $\kappa^2 h^{3/2-\varepsilon}$ sufficiently small, we have shown the claim of Theorem 3.4.

Proof of Theorem 7.6. Let $\xi \in \mathcal{H}_0$ be the solution for $\psi = u - R_h u$. Then, we derive along the lines of [3]

$$\|u - R_h u\|_\mu^2 = a_\kappa(u - R_h u, \xi - I_h \xi)$$

where we used that also $I_h \xi \in \mathcal{H}_0$ and conclude

$$\begin{aligned} \|u - R_h u\|_\mu^2 &\lesssim \left(\mu^{1/2} \gamma^{-1} h^{3/2-\varepsilon} \|u - R_h u\|_\mu + \eta \|u' - (R_h u)'\| h^{1/2-\varepsilon} \right. \\ &\quad \left. + \gamma^{-1/2} h^{1/2-\varepsilon} \gamma^{1/2} \|u' - (R_h u)'\| \right) \gamma \|\vartheta_2\|_{H^{3/2-\varepsilon}} \\ &\lesssim \left(\mu^{1/2} \gamma^{-1} h^{3/2-\varepsilon} \|u - R_h u\|_\mu + \eta \|u' - (R_h u)'\| h^{1/2-\varepsilon} \right. \\ &\quad \left. + \gamma^{-1/2} h^{1/2-\varepsilon} \gamma^{1/2} \|u' - (R_h u)'\| \right) (\mu^{1/2} + \eta) \|u - R_h u\|_\mu \end{aligned}$$

and hence the claim. $\square$

7.1. Necessary modification for the new orthogonality constraint. The main modification is the coercivity of E'' which only helps perpendicular to iu . However, let $\varphi_h \perp I_h iu$. We decompose this as

$$\varphi_h = \widehat{\varphi} + \alpha iu, \quad \widehat{\varphi} \perp iu, \quad \alpha = \frac{(\varphi_h, iu)}{\|u\|^2}.$$

Thus, we compute

$$\|\varphi_h\|_{H_\kappa^1}^2 \lesssim \|\widehat{\varphi}\|_{H_\kappa^1}^2 + \|\alpha iu\|_{H_\kappa^1}^2 \lesssim \rho_\kappa(u) E''(\widehat{\varphi}, \widehat{\varphi}) + \frac{|(\varphi_h, iu)|^2}{\|u\|_{H_\kappa^1}^2}$$

where we used the coercivity and Lemma 2.1 for the norm exchange for u . Further, we obtain

$$|(\varphi_h, iu)| = |(\varphi_h, iu - I_h iu)| \lesssim \|\varphi_h\| \kappa h \|u\|_{H_\kappa^1}$$

and we can absorb it to the left hand side for κh small. Further, since $E''(\widehat{\varphi}, \widehat{\varphi}) = E''(\varphi_h, \varphi_h)$ by the kernel property, we obtain coercivity.

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APPENDIX A. SOME COMPLEX NUMBER ESTIMATES

Lemma A.1. *Let $u, v \in \mathbb{C}$, and $\delta \geq 0$.*

(a) *For $|u|, |v| \geq 1 + \delta$ it holds*

$$\operatorname{Re}\left(\frac{u}{|u|} - \frac{v}{|v|}\right)(u - v)^* \geq (1 + \delta) \left| \frac{u}{|u|} - \frac{v}{|v|} \right|^2$$

(b) *For $|u| \geq 1 + \delta$ and $|v| \leq 1 + \delta$ it holds*

$$\operatorname{Re}\left((1 + \delta)\frac{u}{|u|} - v\right)(u - v)^* \geq \left|(1 + \delta)\frac{u}{|u|} - v\right|^2$$

Proof. (a) We write $u = ra$, $v = sb$ for $|a| = |b| = 1$ and denote by $c = \operatorname{Re} ab^*$. With this, we compute $|a - b|^2 = 2(1 - c)$ and further

$$\operatorname{Re}\left((a - b)(ra - sb)^*\right) = r(1 - c) - s(c - 1) = (r + s)(1 - c) = \frac{r + s}{2}|a - b|^2.$$

Noting that $r + s = |u| + |v| \geq 2(1 + \delta)$ gives the claim.

(b) Again, we write $u = ra$ for $|a| = 1$ and $b = v$, and set $c = \operatorname{Re} ab^*$.

$$|(1 + \delta)a - b|^2 = (1 + \delta)^2 - 2c(1 + \delta) + |b|^2$$

and further

$$\operatorname{Re}\left(((1 + \delta)a - b)(ra - b)^*\right) = r(1 + \delta - c) - c(1 + \delta) + |b|^2$$

Since $|c| \leq 1 + \delta$ and $r \geq 1 + \delta$, this implies

$$\operatorname{Re}\left(((1 + \delta)a - b)(ra - b)^*\right) \geq (1 + \delta)(1 + \delta - c) - c(1 + \delta) + |b|^2 = |(1 + \delta)a - b|^2,$$

and hence the claim. □

APPENDIX B. SOBOLEV EMBEDDING FOR DIMENSION TWO

Although this result is apparently well known in the literature, we did not find a source giving a full proof of it. Thus, we give it here in full detail relying on extension operators and the full space result by Ruf [17].

Lemma B.1. *Let $\Omega \subset \mathbb{R}^2$ be a bounded Lipschitz domain. Then, there is a constant C only depending on the domain Ω such that for all $p \geq 1$ it holds*

$$\|v\|_{L^p(\Omega)} \leq C\sqrt{p}\|v\|_{H^1(\Omega)}.$$

Proof. Let $T: H^1(\Omega) \rightarrow H^1(\mathbb{R}^2)$ be a stable extension operator as in [22, Theorem VI.5]. Then, it is sufficient to prove the result for the full space. In addition, we may assume that the functions vanish sufficiently far away from Ω , say with support in $\widehat{\Omega} \supseteq \Omega$.

We state the result from [17, Theorem 1.1], where there is a constant $K > 0$ such that

$$\sup_{\|v\|_{H^1(\widehat{\Omega})}=1} \int_{\widehat{\Omega}} (e^{4\pi v^2} - 1) dx \leq K,$$

and K is independent of $\widehat{\Omega}$. Hence, we set $\widehat{u} = Tu$, and $v = \frac{\widehat{u}}{\|\widehat{u}\|_{H^1(\widehat{\Omega})}}$, and use that $e^t - 1 \geq \frac{t^m}{m!}$ for all $m \geq 1$. This implies

$$\frac{(4\pi)^m}{m!} \int_{\widehat{\Omega}} \frac{|\widehat{u}|^{2m}}{\|\widehat{u}\|_{H^1(\widehat{\Omega})}^{2m}} \leq K \quad \implies \quad \|\widehat{u}\|_{L^{2m}(\widehat{\Omega})} \leq (m!)^{1/2m} \frac{K^{1/2m}}{\sqrt{4\pi}} \|\widehat{u}\|_{H^1(\widehat{\Omega})}$$

for all natural numbers $m \geq 1$. By Stirling's formula, we finally obtain

$$(B.1) \quad \|\widehat{u}\|_{L^{2m}(\widehat{\Omega})} \leq \widehat{K} \sqrt{2m} \|\widehat{u}\|_{H^1(\widehat{\Omega})}$$

for some constant $\widehat{K}$ only depending on K . Now, for any p with $2m - 2 \leq p \leq 2m$, we apply (B.1) for $m - 1$ and m to obtain the assertion. $\square$

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